

DANIEL R. DUNCAN

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EDUCATION

Master of Science, Watershed Science

Colorado State University, Fort Collins, CO

Expected: May 2027

GPA: 4.0

Bachelor of Science, Physical Geography

Texas State University, San Marcos, TX

May 2018

GPA: 3.19

RESEARCH INTERESTS

- Water quality & streamflow forecasting
- Contaminant fate & transport
- Wireless Mesh Network (WMN) & remote sensing

HYDROLOGY, SUSTAINABILITY, AND CONSERVATION EXPERIENCE

Graduate Research Associate, Radical Open Science Syndicate, Fort Collins, CO

February 2026 – Present

- Maintains a collection of real-time water quality monitoring stations along the Cache la Poudre River; ensures proper calibration and measurements of in-situ sondes.
- Performs data extraction, normalization, and QA/QC for stream flow and water quality datasets; created workflow to extract and normalize 20+ years of data from excel files into parquet format.
- Participates in quarterly stakeholder engagements; develops reports and documentation conveying research findings.
- Explores and develops baseline forecast models for stream water quality.

Marine Ecosystems Specialist, Northwest Straits Commission, Mount Vernon, WA

July 2025 – April 2026

- Planned & coordinated aquatic invasive species (AIS) trapping efforts between private, state, and federal stakeholders within Washington State. Efforts resulted in the removal of 1600+ European green crabs (EGC) during 2025.
- Responsible for the normalization, transformation, analysis, and geoprocessing of all field data; provides accurate tracking of EGC activity, leading to regionally tailored AIS removal strategies.
- Led a multi-agency team in AIS trapping, species identification, field data entry within the Drayton Harbor.
- Conducted statistical analysis of EGC propagation; generates reports for the State of Washington.
- Utilized RStudio and Python to automate data retrieval and conduct data analysis to enhance trapping efforts.

Hydrologist Intern, National Weather Service, Louisville, KY

July 2024 – December 2024

- Utilized Esri products to develop operational Flood Inundation Mapping (FIM) Dashboard, resulting in improved FIM analysis and Impact-Based Decision Support Services (IDSS) workflows for forecasters.
- Provided real-time hydrologic analysis during severe weather events; monitored QPE/QPF, flash flood guidance, and observed river gauge data to ensure timely and accurate flood advisories and warnings.
- Maintained and optimized local ArcGIS Online repository, improving access to and analysis of locally generated data.
- Collaborated with local emergency managers, neighboring WFOs, OHRFC, and public stakeholders to share hydrologic forecast products and services, enhancing community preparedness and response efforts.
- Authored and developed local FIM procedures and a comprehensive training plan, ensuring consistency and accuracy in flood forecasting. Trained forecasters in FIM, enhancing local forecasting capabilities and operational readiness.

PROFESSIONAL EXPERIENCE

Senior Network Assurance Officer, U.S. Army, Fort Knox, KY

July 2023 – February 2025

- Information Systems Security Manager (ISSM) of a logistics organization, leading three multifunctional teams that ensure the physical and network security of 400+ military and government personnel.
- Oversaw the validation of cybersecurity policies, guidelines, and regulations by supporting compliance activities. Enabled organization to receive a “commendable” rating during regulatory cybersecurity inspection.

- Managed the organization's incident response program; successfully mitigated 5+ cyber incidents while coordinating concurrent response lines.
- Developed the organization's cybersecurity program, ensuring federal regulatory compliance. Overhauled organizational procedures and remediated deficiencies of 300+ user accounts.

Data Systems Engineer, U.S. Army, Fort Eisenhower, GA

May 2022 – July 2023

- Advise and brief commanders, stakeholders, and upper management on information systems policy, ensuring informed decision-making and alignment with security objectives.
- Responsible for the design and implementation of network and server infrastructure for a 500+ user organization.
- Configured and deployed Active Directory Domain Services in a multi-forest and MS Exchange environment.
- Utilized PowerShell ISE to perform report automation, bulk user account updates, and security auditing.

Tactical Control Officer, U.S. Army, Baumholder, Germany

May 2018 – April 2022

- Served as supervisor for organization's immediate response force; coordinated short-notice deployment of 90 personnel and 60 vehicles worth \$400 million to the Middle East to provide Air & Missile Defense (AMD) coverage.
- Commanded 60+ personnel during short-notice evacuation of Afghanistan, coordinating with U.S. Air Force security personnel and federal agencies to protect and escort 40,000+ evacuees; awarded Humanitarian Service Medal.
- Planned, resourced, and oversaw the movement of 50+ live missile interceptors across two continents; enabled multiple organizations to maintain combat power during the increase of real-world ballistic missile threats.

Tactical Control Officer, U.S. Army, Osan, South Korea

October 2018 – November 2019

- Ranked as #1 fire control platoon leader within the organization, established an effective 12-week unit training plan, leading to the successful certification of three AMD crews.
- Coordinated joint and multinational AMD operations for Korean theater of operations; served as primary tactical control officer, in charge of technical control and management of the Patriot missile system.
- Managed readiness, maintenance, and accountability of Patriot air defense equipment, valued at over \$120 million.
- Led four cross-functional teams as section supervisor; Responsible for morale, health, welfare, and professional development of over 30 personnel. Conducted performance reviews of four mid-level managers.

RELEVANT COURSEWORK

Undergraduate & Graduate Level

- Land Use Hydrology, Watershed Measurements, Water Resources Planning & Management, Environmental Data Science, Climatology, Oceanography, Meteorology, GIS, Remote Sensing

Technical

- Cisco Network Engineering, Windows Server, LAN/WAN Management, Microsoft SCCM, Active Directory, Database Management, ITIL, RMF, PowerShell, Git, Python, R

HONORS, AWARDS, & CERTIFICATIONS

Meritorious Service Medal – U.S. Army

2025

- Awarded for meritorious service as a Senior Network Assurance Officer and Air Defense Artillery officer during three overseas rotations and one deployment.

Humanitarian Service Medal – U.S. Army

2021

- Awarded for serving directly in Operation Allies Refuge, providing humanitarian relief support to refugees who were evacuated following cessation of U.S. operations in Afghanistan.

Certificate of Special Congressional Recognition

2015

- Awarded for work as a firefighter & Emergency Medical Technician (EMT) during the 2015 Memorial Day flooding within Hays County, Texas and the surrounding areas. Continuous operations lasted for 72+ hours.